FGAI IA 2: Official Assignment Submission

Course: Foundations of Generative AI (ICT4403)

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Assignment Title	FGAI IA 2
Developed System Title	VerseForge: Shakespeare-Style Autoregressive Text Generation Using Recurrent Neural Networks
Dataset Name & Link	**Tiny Shakespeare** https://raw.githubusercontent.com/karpathy/char-rnn/master/data/tinyshakespeare/input.txt
GitHub Repository Link	**https://github.com/Yashraj88888/VerseForge-Shakespeare-Style-Text-Generator**

1. Project Overview & Objectives

VerseForge is a word-level autoregressive neural text-generation system trained on Shakespeare's literary corpus. The objective of this assignment is to develop, train, and empirically evaluate four distinct recurrent neural network architectures under strictly comparable experimental conditions: Simple RNN, Bidirectional RNN, Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU).

2. Dataset Characteristics & Preprocessing Methodology

• **Corpus Statistics:** Tiny Shakespeare (1,115,394 characters; 202,651 raw words; 253,599 total tokens).
• **Tokenization:** Word-level tokenization preserving punctuation marks (. , ! ? ; :) as distinct semantic tokens.
• **Data Partitioning:** Chronological partition (80% Train, 10% Validation, 10% Held-out Test). Chronological splitting strictly avoids lookahead leakage.
• **Vocabulary Construction:** 5,959 unique word tokens fitted solely on the training partition (min_freq = 2). Unseen tokens map to <UNK> and padding maps to <PAD>.
• **Sequence Alignment:** Fixed context window of length 25 words with next-word targets, producing 50,714 training sequence pairs.

3. Recurrent Architectures & Autoregressive Validity

• **Simple RNN:** Embedding (dim=64) → SimpleRNN (units=128) → Dense Softmax (vocab=5,959).
• **Bidirectional RNN:** Embedding (dim=64) → Bidirectional(SimpleRNN, units=64 each) → Dense Softmax.
Autoregressive Guarantee: Bidirectional RNN operates strictly across the historical context [w_1, ..., w_T] without lookahead into future target tokens w_{T+1}.
• **LSTM:** Embedding (dim=64) → LSTM (units=128) → Dense Softmax with constant error carousels.
• **GRU:** Embedding (dim=64) → GRU (units=128) → Dense Softmax with coupled reset and update gates.

4. Quantitative Experimental Comparison

All four architectures were trained with identical hyperparameter configurations: Batch Size = 64, Optimizer = Adam (lr = 0.001), Loss = Sparse Categorical Cross-Entropy, Early Stopping Patience = 3.

Architecture	Test Loss	Perplexity	Accuracy (%)	Training Time	Best Epoch
Simple_RNN	5.2899	198.33	13.07%	64.2s	4
Bidirectional_RNN	5.2002	181.31	12.88%	53.9s	3
LSTM	5.2939	199.12	13.36%	158.6s	6
GRU	5.2530	191.15	13.25%	126.1s	4

5. Sampling Temperature Analysis & Qualitative Lexical Metrics

Sequential generation was performed at temperatures **T = 0.5**, **0.8**, and **1.2** across standardized seed prompts. Lexical diversity was quantified using Distinct-1 (unigram), Distinct-2 (bigram), and repetition ratios:

• **T = 0.5 (Low Entropy):** High local syntax but low diversity (Distinct-1 ≈ 0.25–0.31). Degenerates into repetitive cycles.
• **T = 0.8 (Optimal Balance):** Balanced exploration with natural Shakespearean prose cadence (Distinct-1 ≈ 0.47–0.54, repetition ≤ 0.14).
• **T = 1.2 (High Entropy):** High lexical diversity (Distinct-1 > 0.70) accompanied by syntactic disintegration.

6. Representative Generated Text (Seed: *"To be or not to"*)

GRU (T = 0.8): *"to be or not to least a be enough in night. first senator: then, my come that been his mercy and him my wounds lay. bishop than aumerle: what is though for weeping, catesby! duke elizabeth: o casting! pompey, well, my lord, i"*

7. Conclusion & Architectural Justification

Bidirectional RNN achieved the lowest held-out cross-entropy test loss (5.2002) and perplexity (181.31) due to dense bidirectional context encoding of the 25-token prompt without causal lookahead. Among strictly causal recurrent baselines, GRU demonstrated the strongest trade-off of rapid training (126.1s), low perplexity (191.15), and superior qualitative phrasing over Simple RNN (198.33).